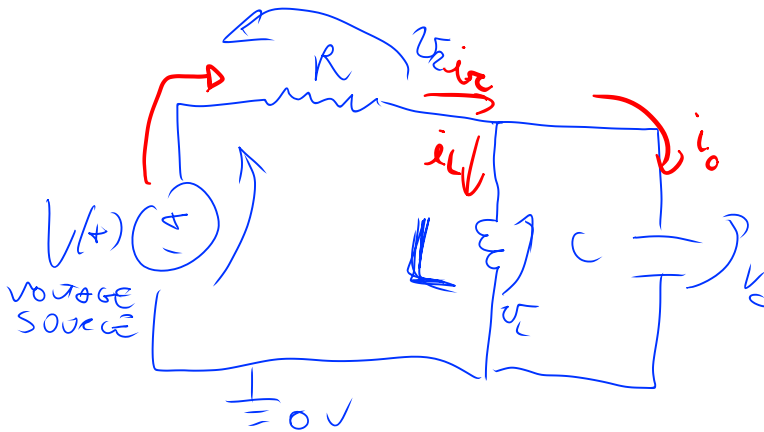


Today, let's take a look at a model of a simple electrical circuit.
Specifically, let's take a look at the RLC circuit.



The arrow indicates which side has a higher potential. R stands for resistance, L is the inductor, and C the condenser.

The control objective: we want to control the voltage across the condenser $v_C(t)$, by acting on $v(t)$, the actuator.

We have some nomenclatures for v , given by v_L and v_R . We also have the current. The current goes through the resistance, being called i_R , then it goes through L being called i_L , and i_C through the condenser.

Let's write the physical equations that govern this system.

Here we have four systems: voltage source, resistance, conductor and inductor.

Let's start from modelling equations of single components.

For the voltage source, as of now we've got nothing to model.

For the resistor, we have that $v_R(t) = R \cdot i_R(t)$.

For the condenser, we have that $i_C(t) = C \cdot \frac{dv_C(t)}{dt}$.

For the inductor, we have that $v_L(t) = L \cdot \frac{di_L(t)}{dt}$.

Then, let's take a look at equations of the network.

One of them is given by Kirchoff law, with $i_R(t) = i_L(t) + i_C(t)$, also known as current equation.

We also see that the inductor and the condenser are connected to the same line, having the same voltage level (cables are at the same voltage levels always, and I mean the part between two components).

Then we also have $v_L(t) = v_C(t)$, as C and L are connected to same cables.

If we consider one point in the lower part, and one above the voltage source, what is the voltage difference? We could measure the voltage difference either going through the shortest path, thus having a voltage difference of v , or through the longest path, so $v(t) = v_C(t) + v_R(t)$.

So, let's put together all the equations.

For step 1, we have Nomenclature, We need to identify u , y and x .

We have that $u \equiv v$, as v is the parameter we are able to control. As for the output, we want to control the voltage of the condenser, so v_C . Thus, $y \equiv v_C$.

What about the state variables? They're given by the variables we differentiate, so v_C and i_L . Thus:

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} v_C \\ i_L \end{pmatrix}$$

The next step is to find the State Space model, so something like this:

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases}$$

The system given by the control and the output, consisting of a single input and single output, takes the name of SISO system (single input, single output).

We need to find $\dot{x}$.

$$\dot{x}_1 = \dot{v}_C = \frac{1}{C} \cdot i_C =$$

Because of the fact that i_C is neither a state variable or a control variable, and we have to write $\dot{x}_1$ in function of u and x , we need to further work on it. So:

$$= \frac{1}{C} \cdot (i_R - \underbrace{i_L}_{x_2}) = \frac{1}{C} \cdot \left(\frac{v_R}{R} - x_2 \right) =$$

v_R is neither a state variable or control variable, so we need to rewrite it.

From the second voltage equation, we have that $v_R = v - v_C$, so:

$$= \frac{1}{C} \cdot \left(\frac{v - v_C}{R} - x_2 \right) =$$

where $v = u$, and $v_C = x_1$, thus:

$$= -\frac{1}{RC} \cdot x_1 - \frac{1}{C} \cdot x_2 + \frac{1}{RC} \cdot u$$

As for $\dot{x}_2$:

$$\dot{x}_2 = \dot{i}_L = \frac{v_L}{L} = \frac{1}{L}x_1 + 0x_2 + 0u$$

As $v_L = v_C$.

Now, let's write the output y . We have $y = v_C$, and as $v_C = x_1$ then $y = 1 \cdot x_1 + 0x_2 + 0u$.

These were the equations in scalar form, now let's write them in Matrix form.

$$\dot{x} = \begin{pmatrix} -\frac{1}{RC} & -\frac{1}{C} \\ \frac{1}{L} & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{RC} \\ 0 \end{pmatrix} \cdot u$$
$$y = (1 \quad 0) \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0u$$

In particular, here we have Linear state space equations, as they have the form of

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

We can see how $f(x, u) = Ax + Bu$, and $g(x, u) = Cx + Du$, and they're both linear equations. Whenever we have additions and multiplications they're linear functions.

A linear system is one having a linear model.

We can also have nonlinear state space models, by having one between f and g not linear. An example is given by a simple non linear model of a pendulum.

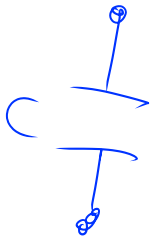
Some notes:

The state space model is an implicit model, it tells us the derivative of the state, as the first equation is given by $\dot{x} = \dots$. To compute not the derivative, but the state, we have to solve the state space equations. The solution is called an explicit model.

≡ Simple Example

Let's focus our attention on the condenser. The equation of the condenser is given by

$$i_C = C \cdot \frac{dv_C(t)}{dt}.$$



Suppose we want to control v_C by acting on the current i_C .

We have that i_C is an implicit model, as we have the derivative. We need to find v_C , and to do that we have to integrate.

$$\frac{dv_C}{dt} = \frac{1}{C} \cdot i_C(t) \implies$$

integral

$$\int_{t_0}^t \frac{dv_C}{dt} dt = \frac{1}{C} \cdot \int_{t_0}^t i_C(t) dt$$

At the left, we have the integral of the derivative, given by: $v_C(t) - v_C(t_0)$.

So:

$$v_C(t) - v_C(t_0) = \frac{1}{C} \cdot \int_{t_0}^t i_C(t) dt$$

So, the solution of the explicit model is given by

$$v_C(t) = v_C(t_0) + \frac{1}{C} \cdot \int_{t_0}^t i_C(t) dt$$

If I want to compute the state of my system at a future time, I need to know the initial state, given by $v_C(t_0)$. The solution thus will be given, in general, by the initial state and the integral of the control.



The key message from this example is this: If we want to know the actual value of the state at a given time, we need to know the initial state of the system.

To recap, an implicit model does not tell us the value of the state, rather of its derivative.

We'll see how to solve a model.

Another quick observation is given by Descriptor Systems.

Many times, physical systems must respect constraints on x and u variables. A Humanoid Robot might have a constraint on the joint angles of an arm, for

example (with minimum and maximum extension).

The constraints can be written as algebraic equations in x and u . In many cases, these constraints are linear. In the linear case, for example, we could have, like in the example above, two state variable and one input.

We might have an example like this: $0 = x_1 - x_2 + 3u$, this being a constraint.

How can we integrate these constraints in our model? They can be integrated into the state equations $\dot{x} = Ax + Bu$.

By integrating them, we get a new model. After $\dot{x}_1$ and $\dot{x}_2$, so, we append our constraints in the case.

We arrive at a model given by

$$E \cdot \dot{x} = \tilde{A}x + \tilde{B}u$$

E has a form of $E = \begin{pmatrix} I \\ 0 \end{pmatrix}$, where I is the identity matrix, and 0 is the matrix of zeroes. We use a matrix of zeroes as it's part of the constraints (imagine we have two more constraints like the one above starting with zeroes).

This model with constraints take the name of Descriptor System.

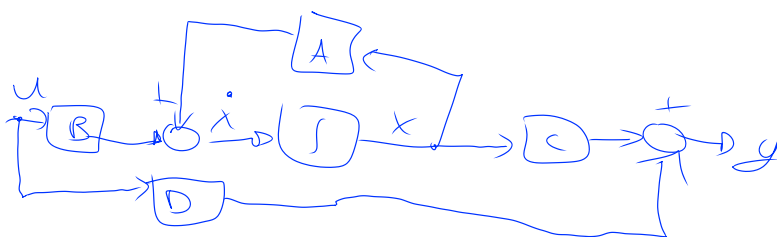
Another note: The state space model is also called a Simulation model. We can use it to build simulators of the system.

Block diagram of the state space equations. Matlab/Simulink works in this way.

Suppose we have a device able to perform numerical integration of a signal.

We are a device given as an input u , and outputting $\int u$.

Let's take this device with input $\dot{x}$, and output $\int \dot{x} = x$.



Then we need to use those equations, and write a block diagram.

When having a variable pass through a block like A, B, C, D , it means we're multiplying it. We also have some sums.

For the $\int$, we need to tell the device what is the initial state.

We can build a simulator for the system. Simulator is fundamental to perform simulations and analysis.

The photo above is valid for a linear system.

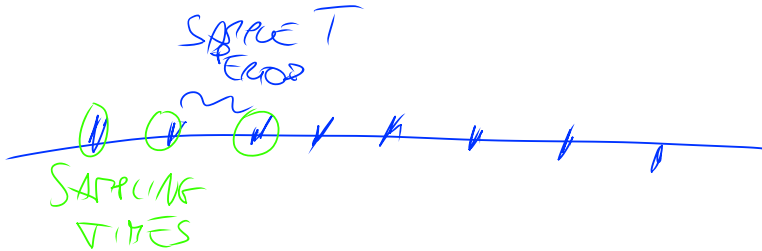
These are examples of continuous times systems (as we have the derivatives).

Physical objects live in continuous time.

Thus, $t \in \mathbb{R}$, with t (time) being a real number (continuous variable, can take any value).

However, the controllers (and computers) work in discrete time.

Thus, we divide time in sampling periods of a given length.



Controls will be taken by controllers at sampling times (end/beginning of the sampling period), quickly computing the control values to be applied in the next slot.

Typically, controls are constant over the sampling period. In the first example, in a sampling period the value of v will be kept constant.

Since computers work in discrete time, they need a discrete time model of the system.

When modelling a physical system from physical equations we find a continuous space model, but to use it with controllers we need to find an equivalent discrete time state space model. This process takes the name of discretization.

Let's take a look at linear case:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

$t \in \mathbb{R}$ denotes a generic time instant.

In the discrete model, we write:

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{cases}$$

$k \in \mathbb{N}$ denotes the generic sampling time/interval.

In continuous time, the first equation tells us the derivative of the state, while in discrete time the first equation tells the next value of the state.

How is discretization performed?

We could approximate the derivative $\dot{x}(t)$ in this way:

$$\dot{x}(t) \cong \frac{x(k+1) - x(k)}{T} \quad \text{when } T \rightarrow 0$$

Then, we take a look at equivalent state space models.

By taking a look back at the example, we picked the following definition of the state:

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} v_C \\ i_L \end{pmatrix}$$

However, this is only one possible choice. Another equivalent choice could be to define the state vectors in another way.

Let's now call the state z . We could have:

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} i_L \\ v_C \end{pmatrix}$$

Clearly, by continuing with this choice, we get different matrices, but the system is the same. For this reason, we say that they're equivalent models, describing the same system.

For one physical objects, we can have infinite many models. How to go from one model to another equivalent one?

State variables are also known as coordinate variables.

Let's write z in the function of x .

We'll have

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

The $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is an invertible matrix. To stress the fact it's invertible, we call it T^{-1} .

Thus, the relation between the two coordinates can be written like this: $z = T^{-1}x$.

Let's find the model in z coordinates.

To do this, we try to find:

$$\begin{cases} \dot{z} = Az + Bu \\ y = Cz + Du \end{cases}$$

To find it, I simply need to compute

$$\dot{z} = T^{-1}\dot{x} = T^{-1}(Ax + Bu)$$

$$y = Cx + Du$$

By having $x = Tz$, we have that

$$Tz = (TT^{-1})x = x \rightarrow x = Tz$$

replacing x with Tz where appropriate.

$$\begin{cases} \dot{z} = T^{-1}ATz + T^{-1}Bu \\ y = CTz + Du \end{cases}$$

Thus, $T^{-1}AT$ will be the $\tilde{A}$ matrix, for example.